Name ____ Date ___________ Period _____

Composition to Verify Inverses

TEACHER KEY | Algebra II Honors | Unit 1 | Day 13 | TEK A2.2.D

Objective: I can compose two functions, use $f(g(x)) = x$ and $g(f(x)) = x$ to prove they are inverses, and state the domain restriction a non-linear pair requires.

Vocabulary

Composition — one function inside another, $f(g(x))$ or $(f \circ g)(x)$. Evaluate from the **inside out**.

The inverse test — f and g are inverses only when $f(g(x)) = x$ AND $g(f(x)) = x$. One direction is not enough.

Order matters — in general $f(g(x))$ **is not** the same as $g(f(x))$.

Part 1 — Warm Up and Prove

Let $f(x) = 3x - 5$ and $g(x) = (x + 5)/3$.

$f(4) =$ **7** $g(7) =$ **4** $f(g(4)) =$ **4** $g(f(4)) =$ **4**

Now prove it in general, both directions.

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$f(g(x)) = 3\cdot(x+5)/3 - 5 = x$ $g(f(x)) = ((3x-5)+5)/3 = x$ Inverses? **yes**

Show that order matters in general: for $f(x) = x + 1$ and $h(x) = 2x$, find both compositions.

$f(h(x)) =$ **$2x + 1$** $h(f(x)) =$ **$2x + 2$** Equal? **no**

Part 2 — A Pair That Looks Right and Is Not

Test $f(x) = 2x + 1$ and $g(x) = 2x - 1$. Compute, do not assume.

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$f(g(x)) =$ **$4x - 1$** Equal to x? **no** Actual inverse of f: **$(x - 1)/2$**

State the lesson in your own words:

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Changing a sign is not undoing a function — you must reverse **every step, in reverse order**.

Part 3 — Non-Linear Pairs and Restrictions

Test $f(x) = x^2$ and $g(x) = \sqrt{x}$. Try $x = 9$ and then $x = -9$.

$f(g(9)) =$ **9** $g(f(9)) =$ **9** $g(f(-9)) =$ **9, not -9**

So $g(f(x)) = x$ holds only when **$x \geq 0$**. In general $g(f(x)) = \sqrt{x^2} =$ **$|x|$**, not x.

Write the complete, correct statement of the inverse relationship for this pair:

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f and g are inverses **only when f is restricted to $x \geq 0$**

Now test $f(x) = x^3$ and $g(x) = \sqrt[3]{x}$ the same way, at $x = -8$.

$g(f(-8)) =$ **-8** Restriction needed? **none** Why the difference from the squaring pair? **cubing is one-to-one**

Part 4 — Error Analysis

A student is given $f(x) = 2(x - 3)^2 + 4$ and claims $f^{-1}(x) = \sqrt{(x - 3)} + 4$.

Test the claim at $x = 6$: the claimed inverse gives **$\sqrt{3} + 4 \approx 5.73$**, and f of that is **not 6**.

Undo the steps in reverse: subtract 4, divide by 2, take the root, add 3.

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Correct: $f^{-1}(x) =$ **$3 \pm \sqrt{((x - 4)/2)}$** Restriction making it a function: **$x \geq 3$ on f**

Where does the ± come from, in terms of Day 12? **f is a parabola, so it is not one-to-one**

Part 5 — Going Further

1. Verify $f(x) = 5x$ and $g(x) = x/5$ are inverses, both directions. **both give x**

2. Show $f(x) = 1/x$ is its own inverse by composing it with itself.

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$f(f(x)) = 1/(1/x) =$ **x** Other self-inverse functions: **x and -x**

3. Prove in general that $f(x) = mx + b$ and $f^{-1}(x) = (x - b)/m$ compose to x.

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4. If $f(g(x)) = x$ but $g(f(x)) \neq x$, can they still be called inverses? Explain.

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No — **both directions are required; a restriction is usually missing**

Exit Ticket

Decide whether $f(x) = 4x - 8$ and $g(x) = (x + 8)/4$ are inverses. Show both compositions, then explain why one direction would not be sufficient proof.

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$f(g(x)) =$ **x** $g(f(x)) =$ **x** Inverses? **yes**